CYBERSECURITY

SSL VPN

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Lab Purpose

The goal of this lab was to simulate a corporate environment which a host on the external network enters a FortiGate webpage that allows the host to remote desktop through an SSL VPN into the internal network encrypted with TLS version 1.2. We also needed to create a proxy which hair pinned itself through the FortiGate firewall and out onto the internet with the firewall serving the purpose of an intermediary connection between the external host and the internet. The external host should be able to remote-desktop into the internal host and access the internet through the configured FortiGate webpage. To do this, split tunneling needs to be configured on the firewall, where traffic with a destination of an internal host and traffic destined for the internet are split, with the host's traffic going to the interior of the network and the internet traffic reaching the internet.

Background Information

There are many different types of VPNs that are used to connect multiple hosts together. Many of these VPN's function in completely different purposes despite seeming to have similar functionalities, namely IPSEC VPNs and SSL VPNs. IPSEC VPNS function most efficiently when used to connect an external host and a network together, giving the host full access to the internal network. SSL VPNs, however, allow an external host to connect to another internal host. While their functionalities seem very similar, the situations in which they are used in are very separate. IPSEC VPNs are used to allow access network-wide to the internal network, an example of this being an external host attempting to access a file system in the internal network or to download/update applications from the internal corporate network. SSL VPNs are usually used to access one specific host, where the external user only has access to that host. An example of this is creating a remote desktop between an exterior user and an interior user to remote desktop into. This external user has no access to anything other than that interior user. Thus, while an IPSEC VPN allows access to the entire corporate network, a SSL VPN will allow access only to one host in the network.

SSL stands for secure socket layer, which operates at the application layer. This means that these hosts can use web browsers to facilitate the sending and receiving of information from this VPN. FortiGate can automatically create a Webpage which a client can use to directly access this VPN connection. These web browsers are called portals, which act as a gateway into the SSL VPN. These portals have bookmarks which specifically link to the intranet but only point to a specific destination. Thus, for instances such like RDP, a portal can contain a bookmark which points towards the destination host. It will list the port number, the IP address, and the host name. If the host has RDP enabled from any destination, then the RDP connection will succeed.

While IPSEC VPNs mainly have site to site VPN capabilities, encrypting all traffic between two networks, SSL VPNs are mainly focused on point-to-point connections. Once a user is signed into the portal, the CA (in this case the firewall) will issue a certificate which will allow any traffic sent between the client and the destination to be encrypted through the encryption method set by the firewall. Usually through TLS or SSL. An encrypted tunnel between the user and the

firewall will then be created and all traffic sent by the user to the firewall will be able to access the internal destination of the intranet.

Another purpose of SSL VPNs are to create hairpins between an external host and the internet. A hairpin is when a VPN server acts as a sort of intermediary between a client and the destination. The client will send the information to the server and the server acts as a proxy. The traffic enters through an interface and gets sent out to the destination from the same interface. This allows all configurations on the server to apply to the traffic being sent, whether it is related to web filters or security measures. It also allows outside traffic to automatically NAT to the internal network. This will allow both the exterior devices and the internal devices to connect. Despite hair pinning being extremely efficient for security and convenience to access the internal network, this can still cause extreme network loads, especially since every piece of traffic that a client sends needs to go through the firewall to reach the internet. This means anything from RDP to a simple internet search.

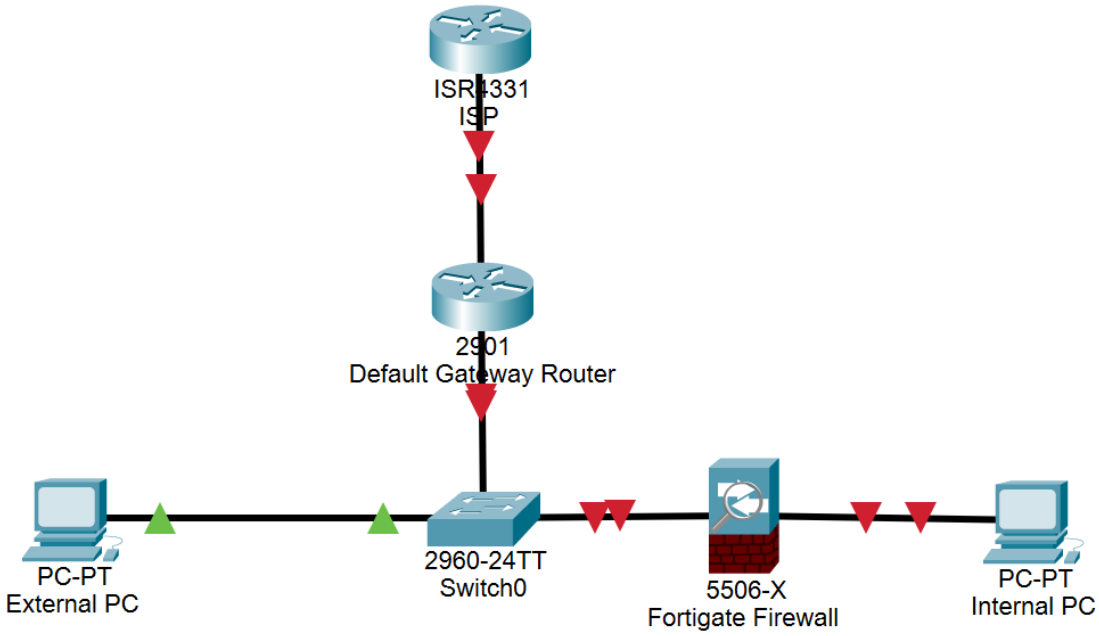
Creating a tunnel that can both RDP and access the internet through a hairpin requires a process known as split tunneling. Split tunneling allows the firewall to split traffic with a destination of the RDP client and traffic destined for the internet. Without split tunneling, all traffic sent will go towards the RDP client, and if the ports are mismatched then the data will not be sent at all.

Lab Summary

Setting up an SSL VPN is extremely straightforward because Fortigate automates many of the steps. We first needed to create a user group under Users & Authentication > User Groups and created a group which will be allowed to access the portal webpage. Then we entered VPN > SSL-VPN Settings and set the VPN to listen on the WAN interface which is connected to the internet. We allowed access from hosts only on the 192.168.40/23 network and allowed access only from the User groups created previously. Then we entered VPN > SSL-VPN portals and allowed split tunneling, which enabled us to allow traffic with a destination of the internet to be forwarded to the internet, whilst traffic with a destination towards the RDP host will be forwarded to the RDP host. To enabled outside traffic to access the web portal, we needed to create two firewall policies. One allowed traffic from the WAN to access the VPN tunnel, and one allowed traffic from the VPN tunnel to access the WAN.

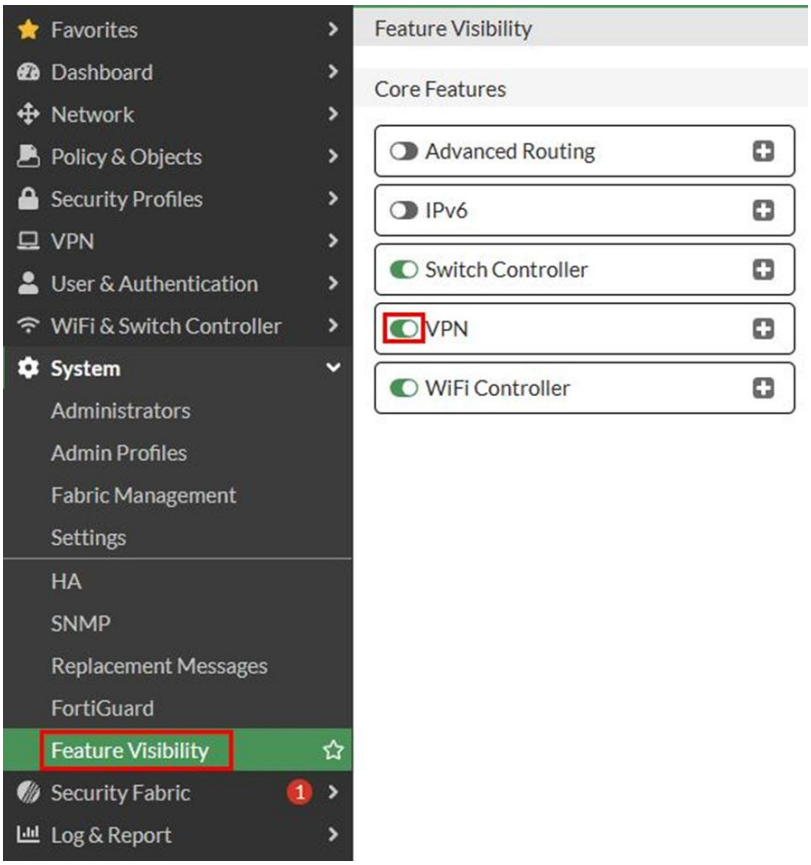
Lab Topology

The External PC can remote desktop through the firewall into the Internal PC through an SSL VPN. It will also hairpin through the Firewall to access the internet using the firewall as a proxy.

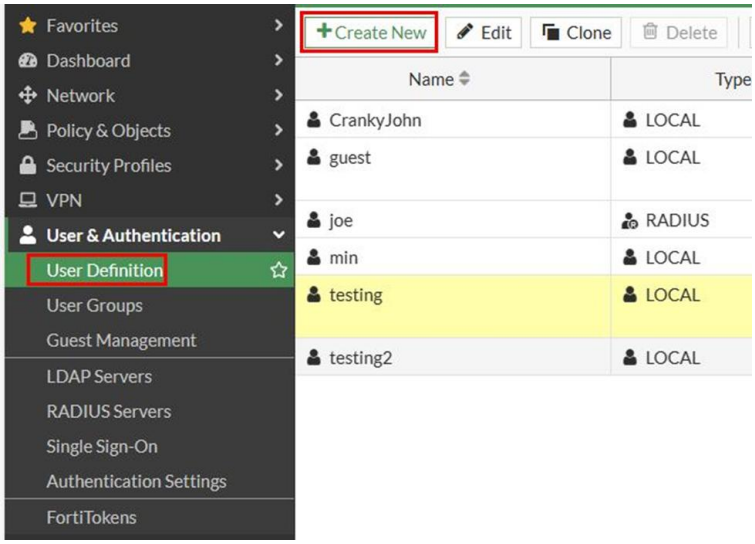


Lab Guide

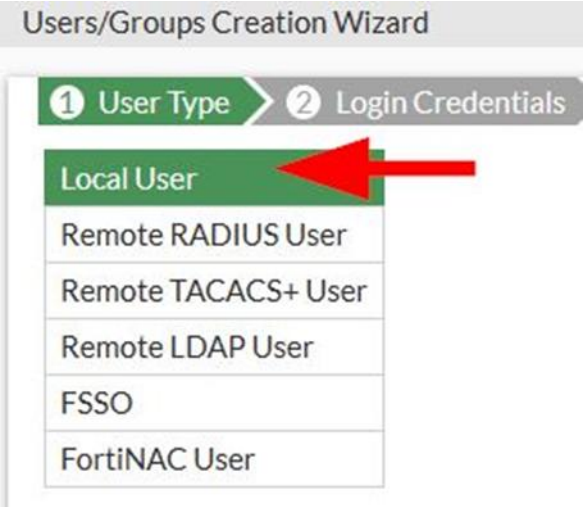
1. Log in to the firewalls GUI, enter oin System > Feature Visibility. Turn on VPN under Core Features to enable VPN capabilities on the firewall.



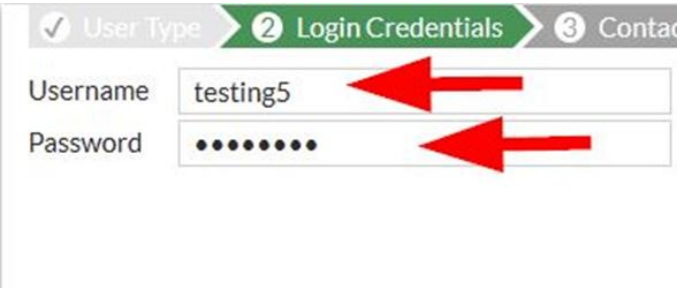
2. Enter User Definition and create a new user to connect to the SSL VPN portal.



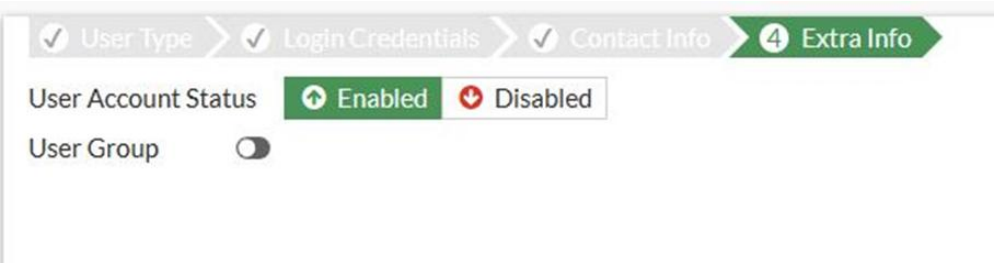
3. Choose local user for this VPN. We are configuring users which will be stored on the local database of the firewall.



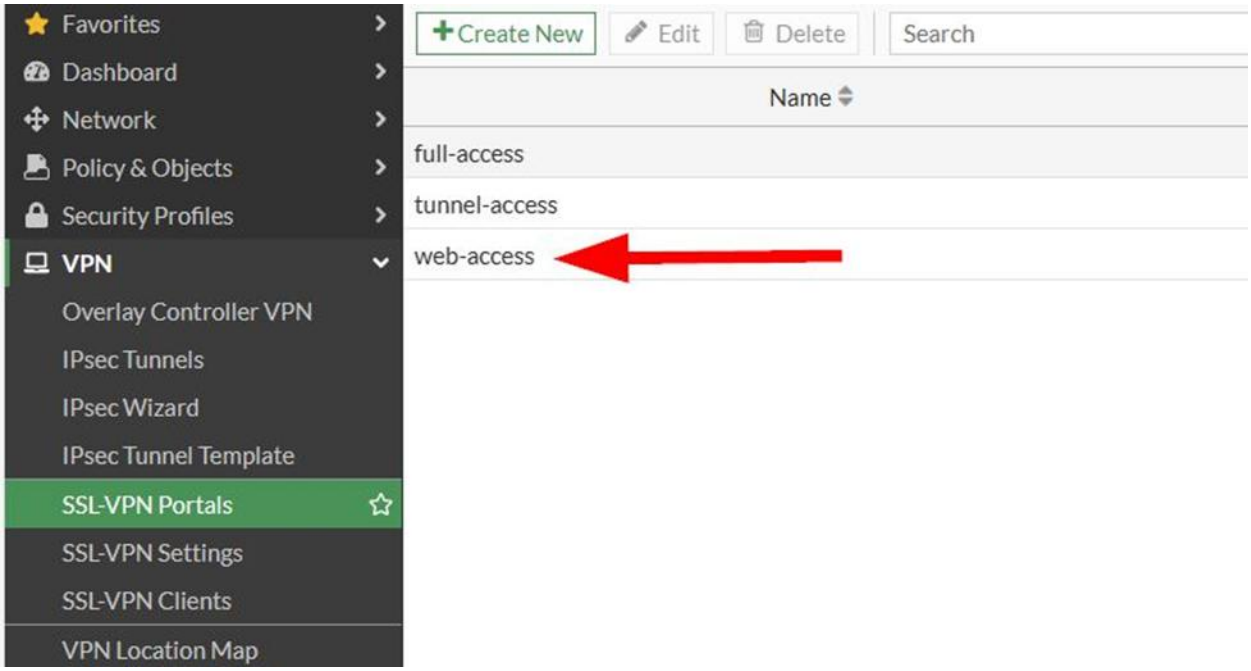
4. Enter the credentials for the user that is going to sign in the SSL VPN portal.



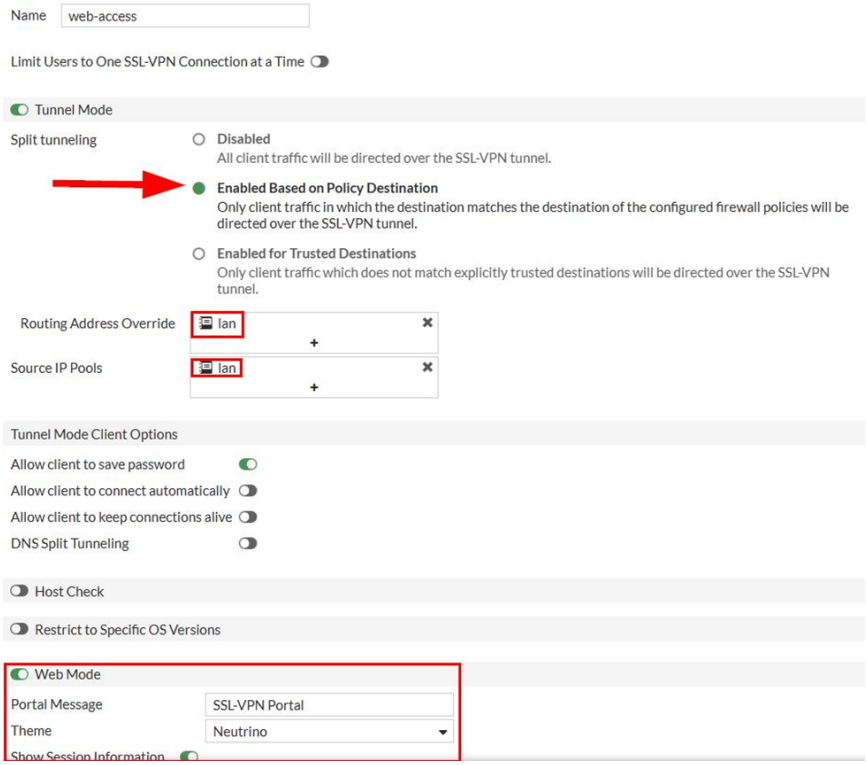
5. Enable the user then finish completion



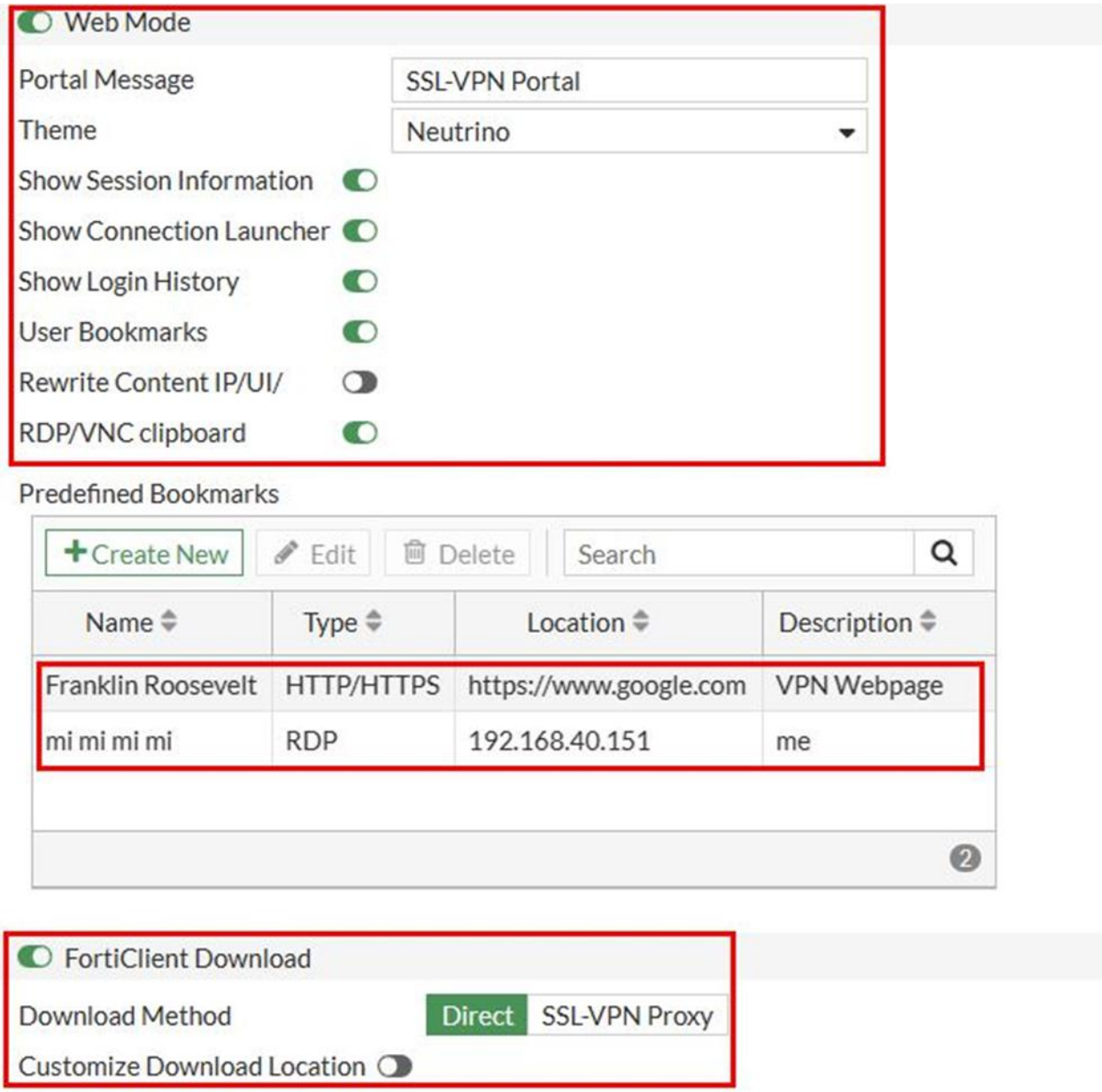
6. Enter VPN > SSL-VPN portals to edit the premade web-access portal.



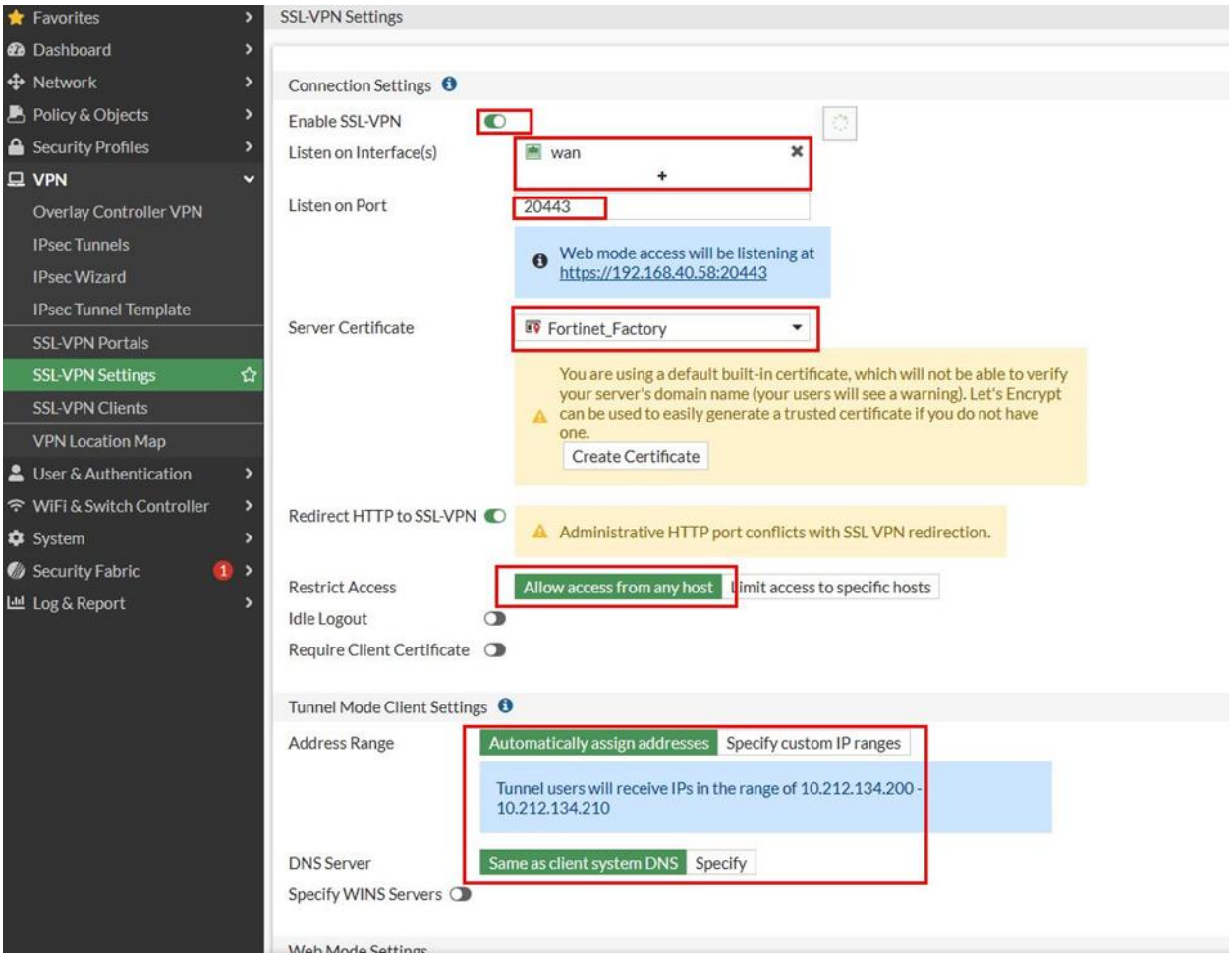
7. Enable tunnel mode and click on Enabled Based on Policy Destination option, this will enable split tunneling so the firewall can forward tunneled traffic to multiple destinations. Set the routing address override and the source IP pool to the LAN's IP address.



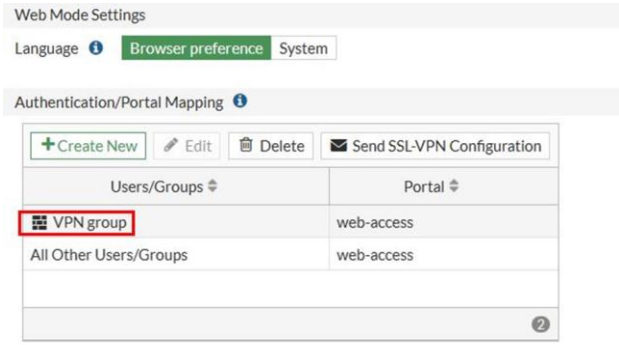
8. Scroll down and edit the Web portal GUI to create bookmarks to allow the client to RDP into the internal network as well as create a bookmark for the client to access google through a hairpin.



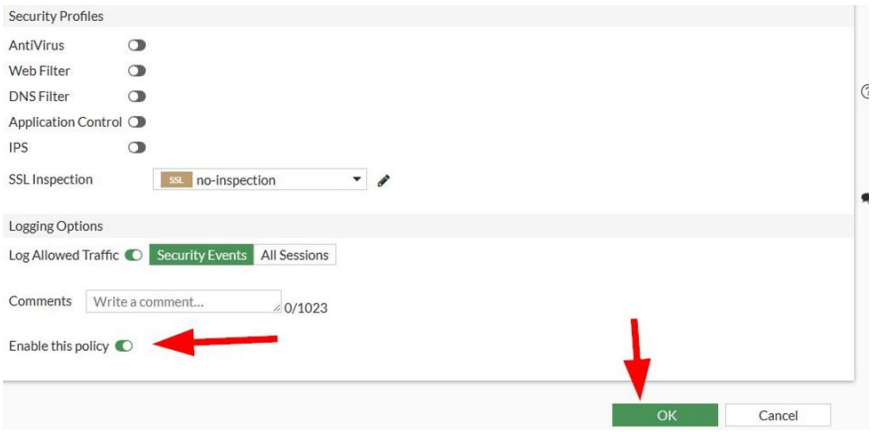
9. Configure the SSL VPN to listen on the WAN port to receive the SSL VPN request from the end user. Use port 20443 or any port related to HTTPS to listen on for the SSL request and set the server certificate to use the default built-in certificate (Fortinet_factory). Set it to automatically assign addresses to the tunnel hosts and allow access from hosts.



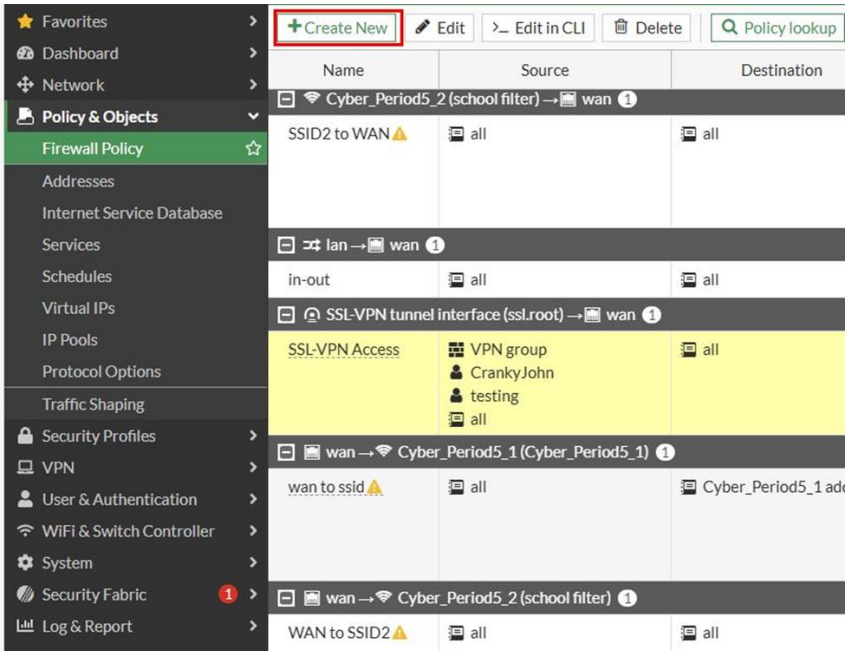
10. In the same menu, scroll down to allow the VPN group previously created for the authentication/portal mapping menu.



11. Scroll down and enable the policy. Click okay to confirm.

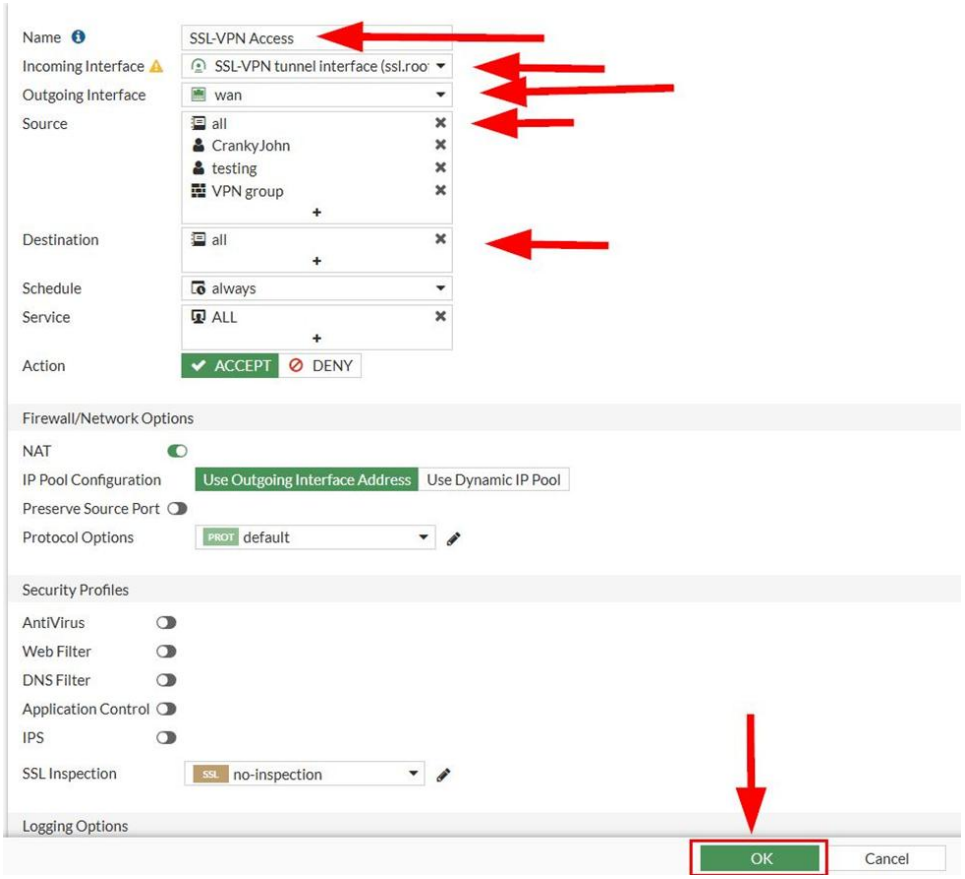


12. Go into firewall policy and create a firewall policy to allow the SSL VPN to accept traffic from the WAN port of the firewall (which usually points towards the internet)



13. Create a name and set the incoming and outgoing interfaces. Under source, allow the users and VPN group that it is set to. Choose where the user which these requests would come from

and pick a destination to allow connections from. Be careful of restricting VPN tunnel access accidentally during this process.



Problems

VPN encryption methods are not compatible with the host's encryption.

> There are a few different encryption methods we could have chosen from when creating the VPN tunnel. At first, the VPN method used resulted in a failed connection because we believe that the encryption methods were incompatible with the hosts. Problem was fixed when we used TLS.

Lab Conclusion

Creating SSL VPNs on any firewall device other than FortiGate would have been much more laborious and difficult. Since FortiGate streamlined most of the process for us whilst setting up the VPN, from auto-NAT when tunneling into the VPN network to automatically creating the FortiGate webpage on the exterior network which allowed access to the firewall. Split tunneling was not challenging at all and neither was creating the web portals because of how automated the deployment was. Overall, in the context of creating SSL VPNs, FortiGate is definitely the best product to allow not only simple deployment, but consistent end to end connection which only goes down if the FortiGate goes down.